

AMAN DADHEECH

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Professional Summary

Manufacturing Systems & Automation Engineer with experience supporting industrial automation, manufacturing process development, production operations, and continuous improvement within high-volume manufacturing environments. Proven track record of improving manufacturability, optimizing production workflows, troubleshooting complex electromechanical systems, and developing scalable manufacturing processes through close collaboration with operations, maintenance, quality, and engineering teams. Passionate about advanced robotics, industrialisation, and bringing innovative products from prototype to production.

Skills

Manufacturing Systems: Process Optimization, Continuous Improvement, Root Cause Analysis (RCA), Production Operations, Manufacturing Analytics, KPI Development, SOP Development, Line Optimization

Industrial Automation & Instrumentation: Industrial Instrumentation & Automation, PLC I/O Verification, 4–20 mA Analog Loops, Pressure, Temperature, Level & Flow Instrumentation, RTDs, Photoelectric Sensors, Proximity Sensors

Engineering & Programming: SolidWorks, AutoCAD, Python, GD&T, Ansys Workbench, Arduino, ESP32

Experience

Oman Sugar Refinery Company LLC

Jul 2024 – Present

Mechatronics Engineer | Manufacturing Systems, Industrial Automation, & Operational Excellence

Sohar, Oman

- Led manufacturing process improvements across packaging and warehouse operations, increasing operational efficiency by 33% through workflow optimization, bottleneck elimination, process standardization, and cross-functional collaboration
- Developed digital manufacturing systems and production reporting tools that improved production visibility, supported operational decision-making, and reduced manual reporting effort by 65%
- Diagnosed manufacturing equipment issues by validating field instrumentation, PLC I/O, sensors, and control systems, minimizing downtime across automated packaging operations
- Led root cause investigations and implemented corrective actions for recurring manufacturing equipment failures, improving reliability and reducing repeat downtime
- Coordinated daily production activities across manufacturing, maintenance, warehouse, quality, and operations teams to improve production flow, equipment readiness, and continuous improvement initiatives
- Developed standardized operating procedures, production documentation, equipment checklists, and operator training material to improve process consistency and operational readiness
- Designed digital production tracking systems integrating manufacturing KPIs, inventory, dispatch, and operational reporting using Excel, SharePoint, and Power Automate

DTU ALTAIR Robotics Lab

Aug 2019 – Jun 2020

Technical Lead

Delhi, India

- Led multidisciplinary teams in competition robotics development, consistently achieving top-10 rankings
- Expanded the robotics organization by 145% through recruitment, onboarding, and mentorship of new members
- Redesigned the inventory and procurement system, reducing variable costs by 13%

Education

Arizona State University

Aug. 2021 – May 2023

Master of Science (M.S.) in Mechatronics, Robotics and Autonomous Systems

Arizona, USA

Delhi Technological University

Jul. 2017 – Jun 2021

Bachelor of Technology (B.Tech.) in Mechanical Engineering

Delhi, India

Engineering Projects

NASA CanSized Satellite (CanSat) | Systems Engineering, Structural Simulation, Mechanical Design

- Designed and developed a deployable delta-wing glider subsystem for the NASA CanSat Competition, contributing to the team's 1st place finish in India and 6th place internationally among 40+ teams
- Engineered a lightweight deployable structural assembly optimized for manufacturability, structural integrity, and cost, reducing bill of materials by 45.7%

Breathing Sensory Feedback | Mechanical Design, Additive Manufacturing, Sensor Integration

- Designed and prototyped a manufacturable low-cost digital spirometer using additive manufacturing, embedded electronics, and sensor integration, reducing projected product cost to approximately 10% of commercial devices
- Collaborated with healthcare professionals to develop a gamified respiratory rehabilitation device, earning Best Project Runner-Up recognition

Foldable Robotics Gripper | Mechanical Design, Numerical Optimization, Rapid Prototyping

- Designed and optimized a bio-inspired robotic gripper with emphasis on lightweight manufacturable structures, rapid prototyping, and numerical optimization, achieving 13 N gripping force